the shredded star’s gas is captured by the black hole and forms an accretion disk, but some of the gas escapes.

In recent years, thanks to improving computer technology, astrophysicists simulated this. Figure 9.5 is from a recent simulation by James Guillochon, Enrico Ramirez-Ruiz, and Daniel Kasen (University of California at Santa Cruz) and Stephan Rosswog (University of Bremen).[22] At time zero (not shown) the star was headed almost precisely toward the black hole and the hole’s tidal gravity was beginning to stretch the star toward the hole and squeeze it from the sides, as in Figure 6.1. Twelve hours later the star is strongly deformed and at the location shown in Figure 9.5. Over the next few hours, it swings around the hole along the blue gravitational-slingshot orbit and deforms further as shown. By twenty-four hours the star is flying apart; its own gravity can no longer hold it together.